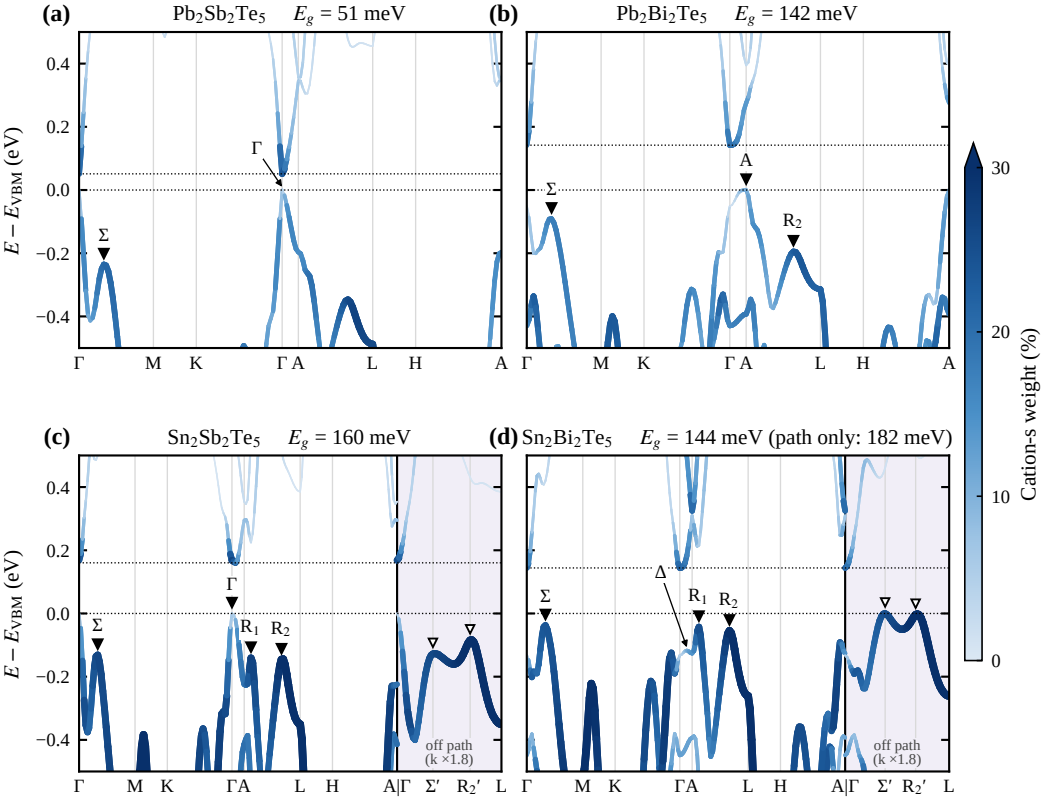

(a)
Pb2Sb2Te5
Eg = 51 meV
(b)
Pb2Bi2Te5
Eg = 142 meV
(c)
Sn2Sb2Te5
Eg = 160 meV
(d)
Sn2Bi2Te5
Eg = 144 meV (path only: 182 meV)
E − EVBM (eV)
Cation-s weight (%)
Γ
Σ
A
R2
Δ
R1
off path
(k ×1.8)
Γ M K Γ A L H A|Γ Σ′ R2′ L